Protein Contact Networks of Antifolate-Resistant *Plasmodium falciparum* Dihydrofolate Reductase

Mid-Semester Project Report

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Abstract

M.Sc. Biological Sciences and B.E. Computer Science (Dual Degree)

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by Ritwik Sethi

Antifolate resistance in *Plasmodium falciparum* arises from point mutations in dihydrofolate reductase that weaken inhibitor binding. The same mutations would be expected to destabilise the enzyme, yet resistant parasites remain viable, so some part of the structure absorbs the cost. Earlier adaptive-walk simulations pointed to a recurring group of residues, the latent compensatory ridge, as the likely absorber, but established only that it is favoured by selection and not that it sits in the right place structurally.

This report covers the first half of a project testing the structural claim. Three one-microsecond trajectories, for the wild type and the S99N and I155L variants, were converted into time-resolved contact networks over ten 100 ns windows, alongside correlation networks and a hybrid of the two, and each construction was checked for how far it can be trusted.

The contact networks are stable: edge counts stay within a nine percent band across a microsecond and vary smoothly with the occupancy threshold rather than collapsing, so the topology is not an artefact of the cutoff. The correlation networks behave differently. The matrix converges at 100 ns, but the thresholded graph swings by a factor of four to nine between windows, which traces to the shape of the correlation distribution rather than to inadequate sampling. The ridge residues form a connected cluster of eight out of eleven, unchanged across all three variants. At the mutation sites the wild type and S99N give identical degrees in every window, which is a result rather than a limitation, because the simulations were all-atom and the backbone was free to respond.

Acknowledgements

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1. Introduction to Problem Statement

1.1 Background

Dihydrofolate reductase reduces dihydrofolate to tetrahydrofolate using NADPH. Tetrahydrofolate supplies the one-carbon units that thymidylate synthase needs, so without a working enzyme a cell cannot make dTMP and cannot replicate its DNA. That is why the enzyme has been a drug target for decades. In *Plasmodium falciparum* it is part of a bifunctional protein, fused to thymidylate synthase in one polypeptide [1], and the antifolates pyrimethamine and cycloguanil act on the reductase domain.

Resistance is well characterised genetically. S108N appears first and causes the initial loss of pyrimethamine affinity, after which N51I, C59R and I164L accumulate on that background to give the highly resistant quadruple mutant [2, 3, 4].

1.2 The Paradox

Why the mutations reduce drug binding is not mysterious. Pyrimethamine and dihydrofolate occupy overlapping space, so a substitution that obstructs one obstructs the other. The puzzle is why the enzyme still works.

Substituting a residue inside a packed active site normally costs something, because side chains are selected over long periods to fit their surroundings. Resistant parasites do not behave like organisms carrying a damaged enzyme. They grow, they transmit, and where drug pressure has been sustained the resistant alleles dominate. Something in the structure takes up the slack.

1.3 The Latent Compensatory Ridge

Earlier work in this group offers a candidate. Adaptive-walk simulations run with the PfPATH framework were started from different resistance mutations and allowed to explore function-preserving substitutions. The walks converged on the same small group of residues regardless of starting point. That group, the latent compensatory ridge, consists of residues 48, 52, 95, 137, 139, 140, 160, 164, 165, 168 and 169 in the numbering used here. Two of them, M95I and F164A, are being examined experimentally by Bis-ANS fluorescence and FTIR.

That result says selection keeps finding these residues. It does not say why.

1.4 Problem Statement

For the ridge to absorb the cost of resistance mutations it has to be in the right place, lying between the mutation sites and the parts of the enzyme that do the chemistry, namely the active site and the NADPH-binding site. If the ridge residues sit on the structural paths connecting those regions, the compensation story has a physical basis. If not, the evolutionary result still holds but needs another explanation.

That is a question about structure, and a protein contact network is a reasonable way to ask it. Representing each residue as a node and each persistent contact as an edge turns the structure into an object graph theory can work on, so questions about paths, bottlenecks and communities become computable.

The project has six objectives in three groups: whether the network holds up across variants and over simulated time, what rewires locally at the active site, the NADPH site and the ridge in each mutant, and whether the ridge is special compared against randomly chosen residue sets and lies on the shortest paths between the resistance sites and the functional core.

1.5 Scope of This Report

This report covers the first half: building the networks and establishing how far they can be trusted. The downstream graph analysis, null models and permutation tests belong to the second half, and two of them wait on input described in Chapter 4.

One note on numbering. The construct is numbered nine residues below the canonical sequence, so S99N here is the literature's S108N and I155L is I164L. The offset was verified residue by residue rather than assumed, and the ridge list above is already in construct numbering.

2. Related Work

2.1 Antifolate Resistance

The genetics were worked out in the late 1980s. Cowman and colleagues [2] and Peterson and colleagues [3] independently identified point mutations in the reductase domain of resistant isolates, with S108N present in essentially every resistant line. Sirawaraporn and colleagues [4] later expressed the mutant enzymes and measured their kinetics, confirming both the loss of pyrimethamine binding and the retention of catalytic activity. The ordering matters here: S108N comes first and the other substitutions accumulate on that background, which is the pattern expected when later mutations depend on earlier ones.

2.2 Protein Structure as a Network

Treating a folded protein as a graph of interacting residues goes back to work on small-world properties in protein structures. Atilgan and colleagues [5] showed that residue networks built from contact maps combine short path lengths with high clustering, and used degree-preserving rewiring as a null model to argue this is a property of the fold rather than of node degrees alone. That rewiring approach, with the backbone chain kept connected, is the null model this project will use.

Most of this literature represents a residue by its Cα atom and draws an edge when two Cα atoms fall within roughly 8 Å. Silveira and colleagues [6] scanned cutoffs systematically and found contact sets become complete near 7.0 Å, which places the conventional 8 Å on the safe side.

2.3 Dynamical Networks from Molecular Dynamics

Static contact maps miss the fact that proteins move. The standard way to bring dynamics in is the approach of Sethi and colleagues [7]: two residues are joined when their heavy atoms come within 4.5 Å for more than 75 percent of frames, and the edge weight derives from the correlation of their motions as $-\log|C_{ij}|$. The arrangement is worth noting carefully, because contacts decide which edges exist and correlation only decides how strong they are. The logarithm turns a correlation into a distance, so strongly coupled residues sit close together and shortest-path algorithms behave sensibly.

The Bio3D package [8] implements this family of methods and supplies the defaults used in much of the applied literature, including a 4.5 Å contact cutoff, a minimum sequence separation of three and a 75 percent persistence requirement. Those defaults inform the parameters in Chapter 3.

Correlation matrices are also used on their own, the motivation being that allosteric communication need not travel along a chain of physical contacts. The difficulty, which Chapter 6 returns to with data, is that turning a correlation matrix into a graph requires a cutoff and there is no principled value for it.

2.4 Where This Work Fits

The prior work established that the ridge is evolutionarily favoured. Convergence in an evolutionary simulation does not by itself show that the residues are structurally positioned to bridge the resistance sites to the functional core. That is a claim about geometry in the folded protein, and it needs testing on structures.

3. Methodology

3.1 Simulation Data

Three systems were simulated for one microsecond each: wild type, S99N and I155L. Each contains 219 residues and 219 Cα atoms, with water retained. Wild type and S99N were written every 20 ps for 50,001 frames; I155L every 10 ps for 100,001 frames. The doubled sampling causes no difficulty because every analysis is defined over a fixed interval of simulated time rather than a fixed number of frames.

3.2 The Pipeline

Figure 3.1 shows the sequence of objects built from each trajectory. Each stage is defined below.

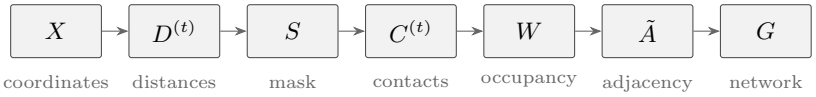


FIGURE 3.1: Construction pipeline. One graph is produced per variant per 100 ns window, thirty in total. After this point the trajectories are not used again and all further analysis is graph theory.

3.3 Coordinates, Distance and Contact

Trajectories were read with MDTraj using chunked loading, selecting the 219 Cα atoms at load time so only those coordinates are ever held in memory. The result is an array X of shape $T \times N \times 3$ with $N = 219$, converted from nanometres to Ångstroms. Representing each residue by one atom reduces about 3,650 protein atoms per frame to 219, which is what makes frame-by-frame analysis of a microsecond tractable. The cost is that side chain identity does not enter the network, a point Chapter 6 returns to.

For every frame the pairwise distance matrix is $D_{ij}^{(t)} = \|X_{t,i,:} - X_{t,j,:}\|_2$.

Residues adjacent in sequence are close because of backbone geometry rather than the fold. Consecutive Cα atoms sit about 3.8 Å apart, and within an α-helix the i to $i+2$ separation is roughly 5.4 Å and i to $i+3$ roughly 5.0 Å. Including them would record secondary structure rather than tertiary contacts and would inflate clustering coefficient and modularity, both of which the project measures. A mask removes them, and a pair is in contact when the distance falls inside a band and the mask allows it:

$$S_{ij} = \mathbf{1}\big[\,|i-j| > s\,\big], \qquad C_{ij}^{(t)} = \mathbf{1}\big[\,d_{\mathrm{lo}} \le D_{ij}^{(t)} \le d_{\mathrm{hi}}\,\big] \cdot S_{ij} \tag{3.1}$$

with $s = 2$, $d_{\mathrm{lo}} = 4.0$ Å and $d_{\mathrm{hi}} = 8.0$ Å.

3.4 Occupancy and the Network

A contact in a single frame may be thermal motion. What matters is the fraction of the window a pair spends in contact:

$$W_{ij} = \frac{1}{T}\sum_{t=1}^{T} C_{ij}^{(t)} \in [0,1] \tag{3.2}$$

The order of operations is important and easy to reverse by mistake. Contacts are evaluated per frame and then averaged, never the other way round, because the contact test is not linear in position and an averaged structure can place an atom where it never was.

Each trajectory is divided into ten windows of 100 ns, giving one occupancy matrix per window. A single time average would hide exactly the variation the analysis is meant to detect. Thresholding gives the network:

$$G = (V, E, w), \qquad V = \{1, \ldots, 219\}, \qquad E = \big\{\,\{i,j\} : W_{ij} \ge \theta\,\big\} \tag{3.3}$$

Each edge carries two attributes. The first, `weight`, holds the occupancy, which is a strength. The second, `distance`, holds $-\log$ of it. The distinction is necessary because several NetworkX routines, betweenness among them, read an attribute named `weight` as a cost to minimise, so passing a strength inverts the meaning and the most persistent contacts come out as the furthest apart. All path-based measures use `distance`.

3.5 Correlation and Hybrid Networks

The dynamic cross-correlation matrix asks whether residues move together rather than whether they touch. Taking fluctuations about each residue's own mean position, $\Delta r_{t,i,:} = X_{t,i,:} - \langle X_{\cdot,i,:} \rangle_t$, and writing $\Sigma_{ij} = \sum_t \Delta r_{t,i,:} \cdot \Delta r_{t,j,:}$,

$$R_{ij} = \frac{\Sigma_{ij}}{\sqrt{\Sigma_{ii}\,\Sigma_{jj}}} \in [-1, 1] \tag{3.4}$$

This is the $N \times N$ scalar matrix, a different object from the $3N \times 3N$ covariance diagonalised in principal component analysis. Edges are drawn where $|R_{ij}|$ exceeds a cutoff c, with the signed value kept separately since anticorrelated motion carries meaning of its own. The hybrid network is the intersection of both criteria:

$$E_h = \big\{\, \{i,j\} : W_{ij} \geq \theta_h \;\wedge\; |R_{ij}| \geq c \,\big\}, \qquad w_h = \sqrt{W_{ij} \cdot |R_{ij}|} \tag{3.5}$$

3.6 Parameters and Software

Table 3.1 lists the parameters. The 8 Å bound and sequence separation of two follow standard practice [6, 8]; $\theta_h = 0.75$ matches the frame criterion of Sethi and colleagues [7]; the correlation cutoff is scanned rather than fixed, for reasons given in Chapter 6.

Table 3.1: Parameters used in network construction.

Parameter	Value	Basis
Node representation	Cα, one per residue	Standard for residue networks
d_{hi}	8.0 Å	Standard upper cutoff [6]
d_{lo}	4.0 Å	Excludes trivially adjacent backbone pairs
s	2	Matches Bio3D `scut` = 3 [8]
θ	0.50	From the sweep in Chapter 6
θ_h	0.75	Sethi *et al.* frame criterion [7]
c	0.3, 0.5	Scanned, not fixed
Window	100 ns	Ten windows per microsecond

The analysis runs in Python, with MDTraj for trajectory reading, NumPy for the distance and correlation arithmetic and NetworkX for the graphs. GROMACS 2026.3 handled trajectory fitting and RMSD. Everything is collected in one notebook so that any variant can be rebuilt in a single pass.

4. Challenges

4.1 An Unsatisfiable Threshold Rule

The first attempt to choose the occupancy threshold automatically used the rule: take the strictest threshold at which the network stays fully connected and all eleven ridge residues are present. No threshold satisfies it, since full connectivity is unreachable at any value and even at 0.25 the giant component tops out at 207 of 219 nodes. The function returned nothing and, unguarded, that propagated into graph construction and produced a type error several steps later.

The real problem was the rule. Isolated residues are not a failure: with sequence separation above two and a 4 to 8 Å band, a surface residue in an extended loop can genuinely have no qualifying partner. The threshold is now chosen from an empirical sweep and isolates are reported rather than eliminated.

4.2 Dependency Conflicts

Installing MDTraj upgraded NumPy from 1.26 to 2.x, breaking the declared pin in `correlationplus`. Rather than rolling back I tested whether the conflict was real by importing each affected package. All of them, including `correlationplus`, Numba and MDAnalysis, worked under the newer NumPy, so the conflict was declarative only and the environment was left alone.

4.3 Open Dependencies

Two items are blocked on information rather than work, and both affect the third group of objectives. The first is the residue lists for the active site and the NADPH-binding site, without which the shortest-path analysis cannot be posed; everything else it needs is in place. The second is confirmation of the numbering convention used by the published ridge list. The check above indicates construct numbering, but that is an inference from a single residue match and should be confirmed against the original methods.

4.4 A Structural Limitation

The second objective asks which contacts are gained and lost at the mutation sites, and a Cα network cannot answer it. The reason is worth stating precisely because it is easy to state wrongly.

The simulations were all-atom. Every side chain was present throughout, and reduction to Cα positions happened afterwards at analysis, so the Cα positions were free to respond and the network is not blind to side chain changes in principle. What it cannot represent is the chemistry itself: an edge records Cα to Cα proximity, so a hydrogen bond a serine hydroxyl makes and an asparagine amide does not is invisible unless it moves a backbone. The network measures consequences rather than causes.

Answering the objective properly means rebuilding on heavy atoms, roughly 1,700 atoms per frame instead of 219 and around sixty times the computation. That is a scoping decision to take deliberately rather than discover halfway through.

5. Experiments

5.1 Equilibration

Before building any network I checked the trajectories were usable, computing backbone RMSD for each variant against frame zero of its own trajectory.

The wild type settles near 0.30 nm from about 20 ns to 320 ns, steps up to roughly 0.42 nm between 320 and 400 ns, and holds that for the rest of the run. S99N reaches 0.45 nm by 50 ns, overshoots to 0.5 nm until about 150 ns, then stays between 0.35 and 0.45 nm through to 950 ns with a late upturn over the final 50 ns. I155L is the least settled, moving between states throughout with excursions towards 0.6 nm near 490 ns and again near 700 ns. All three converge to roughly 0.40 to 0.45 nm after about 400 ns.

Two features matter later: the wild type transition between 320 and 400 ns and the late upturn in S99N, both of which reappear independently in Chapter 6. RMSD was computed on backbone atoms, since side chain motion is noise when the question is whether the structure has plateaued.

5.2 Windowing and Construction

Each trajectory was divided into ten consecutive 100 ns windows, applied identically across variants. The first four sit in territory where the trajectories were still settling. They are kept because they are informative about equilibration, but they are not strictly equivalent to the later windows.

Cα coordinates were extracted once per variant and cached, taking around 140 seconds for the 50,001-frame trajectories. Occupancy matrices were computed for all ten windows by looping over frames, applying the band and mask, and accumulating, and networks were built from these at the chosen threshold. Correlation matrices were computed for the same windows, correlation networks built at cutoffs of 0.3 and 0.5, and hybrids from the intersection of both criteria.

5.3 Robustness Checks

Four checks were run, and they form the bulk of what this report has to report.

Occupancy distribution. The distribution of non-zero occupancy values was examined per variant, to find where the threshold sits relative to the data rather than assuming the choice is harmless.

Threshold sweep. The threshold was swept from 0.90 down to 0.25 in steps of 0.05, recording edge count, largest connected component, isolated residues, and how many ridge residues and mutation sites keep an edge. A network whose properties change smoothly with the threshold is one whose topology is not an artefact of the cutoff.

Correlation convergence. Correlation estimated from molecular dynamics is vulnerable to a specific problem: consecutive frames are highly autocorrelated, so a window may hold thousands of frames while containing far fewer independent samples, and the matrix can look confident while being largely noise. Each window was split in half, a matrix computed from each half, and the two correlated. Both halves come from the same window, so what differs between them is estimation error rather than physics. This was compared against the correlation between adjacent windows, which contains estimation error plus real conformational change. Because each half holds only half the data the two are not directly comparable, so the half-and-half figure was converted to a full-window reliability by the Spearman-Brown relation $r_{\text{full}} = 2r/(1 + r)$. That gives the value adjacent windows would reach if nothing had physically changed, and the shortfall below it measures real change.

Correlation cutoff scan. Cutoffs from 0.3 to 0.8 were scanned, recording edge count, density, mean degree and negatively correlated edges, together with the full distribution of correlation values, to find whether any value is distinguished by the data in the way the occupancy threshold turned out to be.

5.4 Diagnostics

Two further measurements were made. The first takes the subgraph induced by the eleven ridge residues in each window and records its internal edges and largest connected component. The second records the degree

of each mutation site per window and its neighbours. Both are descriptive: a measurement becomes a result only once compared against what randomly chosen residues produce, which belongs to the second half.

6. Results and Discussion

6.1 Summary

Table 6.1 collects the main quantities. The contrast that organises this chapter is between the contact rows and the correlation rows. The contact network moves by nine percent across a full microsecond. The correlation network, built from the same trajectories over the same windows, moves by a factor of three to four.

TABLE 6.1: Main measurements across the three variants.

Quantity	WT	S99N	I155L
Contact edges, range over ten windows	511–557	518–535	511–547
Giant component, range (of 219)	197–215	203–211	200–211
Contact edges, window 5	545	530	520
Correlation edges at $\lvert R\rvert \geq 0.3$, window 5	3,823	2,377	6,044
Correlation edges, swing across windows	4.4×	4.3×	2.9×
Hybrid edges, window 5	455	462	440
Half-and-half correlation	0.77	0.86	0.84
Ridge internal edges	9	9	8–9
Degree of residues 99 / 155	6 / 7	6 / 7	6 to 9

6.2 Occupancy Is Bimodal

Figure 6.1 shows the occupancy matrices for window one with the distribution of non-zero values. The distribution has two sharp modes, one near zero and one near one, each holding more than three hundred residue pairs, with a plateau between them containing about five pairs per bin. Contacts in this protein are either permanent or effectively absent; intermittent contacts barely occur.

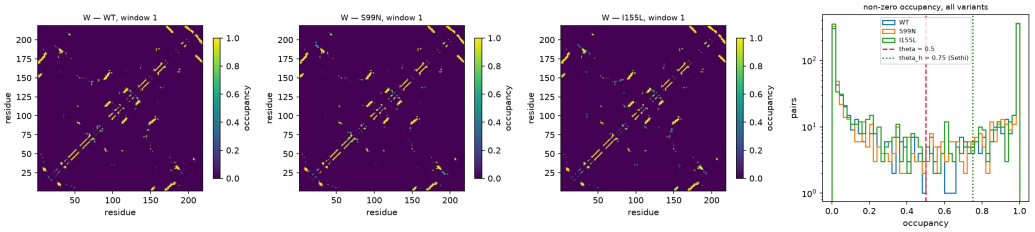


FIGURE 6.1: Occupancy matrices for window one of each variant, with the distribution of non-zero values at right.

The threshold therefore cuts through an almost empty region, so the network is robust to it by construction rather than because a fortunate value was chosen. That is a stronger position than the usual argument from a plateau in the edge count.

One consequence follows and should be recorded. Because surviving edges nearly all have occupancy close to one, the weighted adjacency is close to uniform, so weighted and unweighted centralities will agree and must not later be presented as independent measurements.

6.3 The Threshold Does Not Matter Much

Figure 6.2 confirms what the distribution predicts. Across the sweep from 0.25 to 0.90 the edge count falls smoothly from about 580 to about 400 with no inflection. The giant component drifts from roughly 207 down to 180 of 219 nodes, shrinking without collapsing. All eleven ridge residues keep at least one edge at every threshold, in all three variants.

A threshold of 0.75 is therefore affordable, keeping around 460 edges, a giant component of 189 to 203 and the full ridge. Moving there would align the working threshold with θ_h so that one criterion covers both. All results here were produced at 0.50 and the rebuild is listed in the next steps.

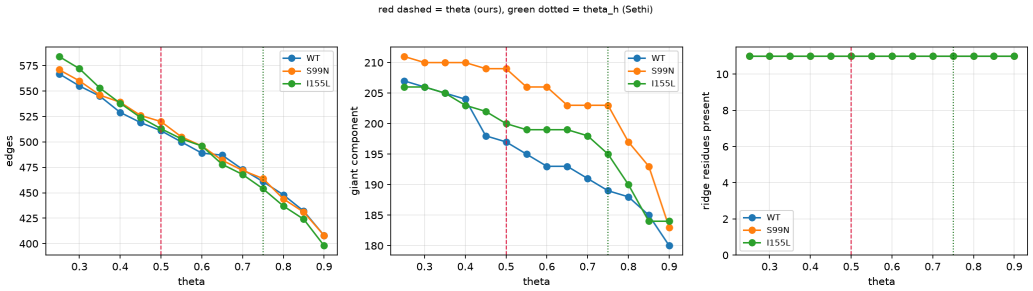


FIGURE 6.2: Edge count, giant component and ridge coverage against the occupancy threshold. Red dashed marks the working value of 0.50, green dotted 0.75.

6.4 The Contact Network Is Stable and Agrees with RMSD

Edge counts stay within 511 to 557 for every variant and window, a range of about nine percent (Figure 6.3). A measurement that wandered by a factor of two within a single trajectory could not support any comparison between variants.

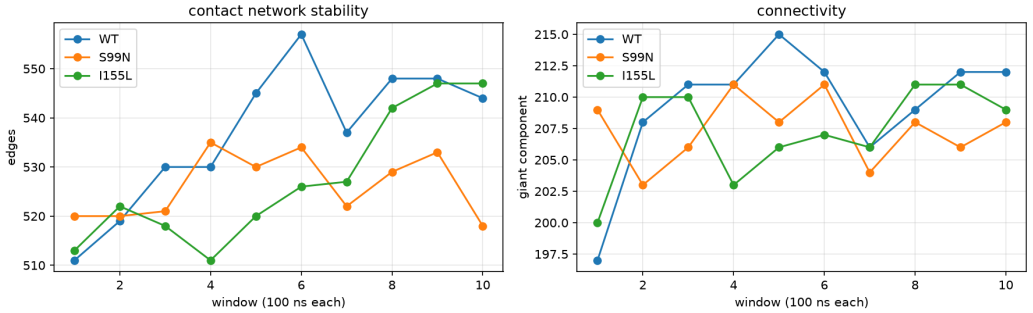


FIGURE 6.3: Edge count and giant component across the ten windows for each variant.

The wild type drifts upward from 511 to about 548 edges. S99N is flat through the middle and drops from 533 to 518 in the final window. I155L climbs from 513 to 547.

The wild type drift coincides with the RMSD step between 320 and 400 ns: the structure settles into a more compact state and picks up persistent contacts. The S99N drop coincides with the late RMSD upturn over the final 50 ns. These agreements matter because the two measurements share no arithmetic. RMSD is geometric and computed from backbone coordinates against a reference; edge count is topological and computed from a thresholded contact criterion. When both flag the same fifty nanoseconds, that is a property of the trajectory rather than of either method.

Between four and twenty residues are isolated at any time, which is expected rather than a problem, for the reasons given in Chapter 4.

6.5 The Correlation Matrix Converges

Within a window, the two halves give matrices agreeing at $r = 0.77$ for wild type, 0.86 for S99N and 0.84 for I155L (Figure 6.4). The Spearman-Brown correction gives full-window reliabilities of 0.87, 0.92 and 0.91.

Those reliabilities are what adjacent windows would reach if nothing had changed physically. Observed adjacent-window correlations sit below them, and the shortfall measures real conformational change: about 0.05 for wild type, 0.06 for S99N and 0.15 for I155L. I155L moves roughly three times as much between windows as either other variant.

That is a result rather than a formality, and it is the third independent indication of the same thing. I155L also has the widest spread of correlation values, is the only variant whose mutation site degrees change, and has the least settled RMSD.

One caveat applies. The split is contiguous, so slow drift within a window depresses the half-and-half figure and mixes estimation error with real change, making the reliability estimates conservative. A split interleaving blocks of about one nanosecond would separate the two properly. Splitting on alternate frames would be wrong

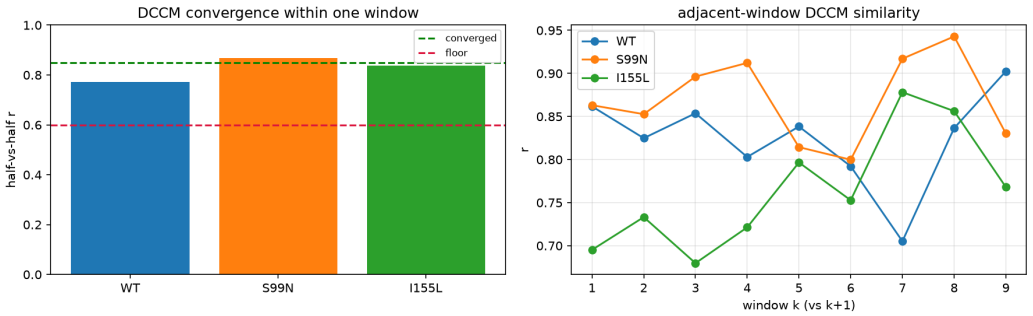


FIGURE 6.4: Left, agreement between matrices from the two halves of one window. Right, agreement between adjacent windows.

in the other direction, since consecutive frames are nearly identical and would report almost perfect reliability regardless.

A reliability of 0.87 to 0.92 means roughly a tenth of the matrix structure does not reproduce. Broad block structure survives that comfortably; individual values near a cutoff do not, which is the subject of the next section.

6.6 The Correlation Network Does Not Converge

The matrix is reproducible but the thresholded graph is not. Figure 6.5 shows edge counts swinging from 2,300 to 10,100 across windows at a cutoff of 0.3, and from 430 to 3,950 at 0.5, against the nine percent range of the contact network over the same trajectories.

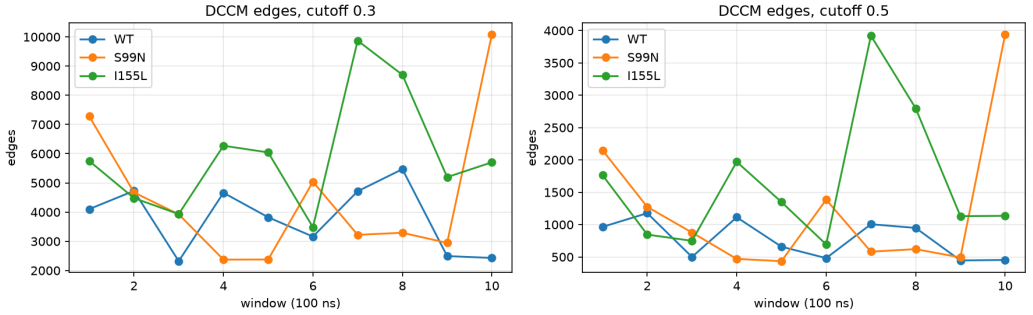


FIGURE 6.5: Correlation network edge counts across windows at cutoffs of 0.3 and 0.5.

These findings look contradictory and are not. The correlation distribution is a single smooth unimodal hill centred near zero, with no gap, no second mode and no natural break. A cutoff at 0.3 passes through its dense shoulder, where several hundred pairs sit in every bin, so a small and physically unremarkable shift in overall coupling moves thousands of pairs across the line. The instability lies in the sensitivity of the edge count to the cutoff, not in the variance of the correlations.

The contrast with the occupancy case is exact. Occupancy is bimodal, so its threshold sits in an empty valley and the network is robust. Correlation is unimodal, so its cutoff sits in crowded territory and the network is fragile. The same operation on two differently shaped distributions gives opposite outcomes.

This is a diagnosis rather than a failure, and it suggests three remedies in increasing order of defensibility: fix the edge density by taking the strongest N edges per window so that every graph is the same size; abandon thresholding and run weighted measures directly on $|R|$; or adopt the arrangement of Sethi and colleagues [7] in which contacts determine the edge set and correlation only the weight, which is what the hybrid already does.

6.7 The Hybrid Is Mostly Contacts

In window five the occupancy networks have 545, 530 and 520 edges, the correlation networks 3,823, 2,377 and 6,044, and the hybrids 455, 462 and 440. The comparison that matters is between S99N and I155L: their

correlation networks differ by a factor of 2.5 and their hybrids by under five percent, so the instability does not propagate.

The hybrid retains 83 to 87 percent of contact edges, meaning the correlation criterion removes very little. That is unsurprising once stated, since residues held between 4 and 8 Å for an entire window are mechanically coupled almost by definition.

The honest consequence is that the hybrid is not an independent line of evidence. It is close to the occupancy network with its edges reweighted, and presenting it as a third corroborating network would overstate what it contributes. Its value is that it is the construction closest to the established method, which makes it the defensible choice for the centrality analysis.

6.8 The Ridge Forms a Persistent Cluster

The eleven ridge residues form nine internal edges with a largest connected component of eight, in nearly every window of every variant. Wild type and S99N are identical throughout; I155L drops to eight edges and seven residues in four of the ten windows. The ridge is therefore neither eleven scattered residues nor a clique, but a connected core of eight with a consistent satellite. The invariance across variants is the substantive observation: whatever the ridge is structurally, neither mutation disturbs it.

This is not yet evidence that the ridge is special. The ridge residues were identified as network hubs in the first place, so finding them clustered and central is partly the original definition reasserting itself. The circularity has to be tested rather than argued away.

The obvious test is a permutation test against random residue sets of the same size, but the obvious version does not work. Drawing the sets uniformly would give a significant result that means nothing, since the comparison would amount to asking whether hubs are hub-like. The null must be built from sets matched on degree, so the only thing distinguishing the ridge is where its residues sit rather than how well connected they are. A second control of high-degree non-ridge residues would strengthen it further.

The claim available on this evidence is that the ridge behaves as a coherent structural unit, not that it is special.

6.9 The Mutation Sites

Residue 99 has degree six in both wild type and S99N in all ten windows, and residue 155 has degree seven in both. The values are identical rather than merely similar. I155L is the only variant where anything moves, with both sites climbing to eight or nine after about the fourth window (Figure 6.6).

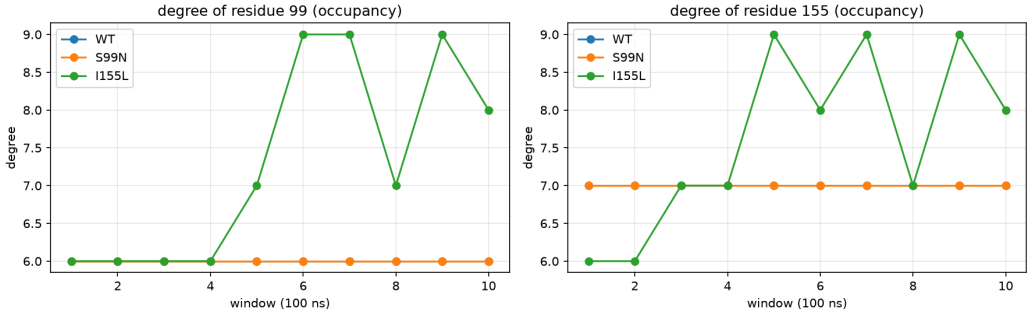


FIGURE 6.6: Degree of residues 99 and 155 across windows. The wild type and S99N traces coincide exactly.

It would be easy to dismiss this as an artefact of coarse-graining, on the grounds that swapping a serine hydroxyl for an asparagine amide does not move the Cα. That reasoning would apply to a coarse-grained simulation, and these were all-atom. The asparagine side chain was present for the entire microsecond, packing against its neighbours and exerting force on the backbone, and the reduction happened afterwards at analysis. The Cα positions were free to respond.

They did not. Read directly, S99N is locally silent at the backbone level: the substitution is absorbed without rearranging which residues the site sits against. That is precisely the tolerance the project exists to explain.

One check remains before stating it in those words. Equal degree is not an equal neighbour set, since residue 99 could have six partners in both variants and have exchanged one. The neighbour lists have been computed and need reading.

I155L differs because isoleucine to leucine changes side chain branching rather than chemistry, repacking the local backbone so the Cα genuinely moves. The timing follows the RMSD excursions near 490 and 700 ns.

6.10 What Would Settle the Compensation Question

Nothing here demonstrates that the ridge compensates. Compensation is a causal and energetic claim, and topology can show that residues are positioned to play a role without showing that they do.

Positioning comes first: whether shortest paths from the mutation sites to the active site and NADPH site run through ridge residues, which is ready to run once the residue lists arrive. Specificity follows through the degree-matched permutation test. Necessity would come from removing ridge residues and measuring whether communication degrades more than when degree-matched random sets are removed.

Beyond that the question leaves graph theory. Energetic calculations, MM/PBSA or free energy perturbation, are what would measure whether a destabilising substitution is offset. The strictest form is epistasis, since compensation means non-additivity: if the free energy change of a resistance mutation combined with a ridge mutation is not the sum of the two separately, the sites interact. No network measurement can establish that, which is where the ongoing Bis-ANS and FTIR work becomes decisive.

Networks nominate candidates rather than establishing causation. The value of the work reported here is that the networks doing the nominating have been shown to be stable, threshold-robust and well enough understood to say which of their outputs can be trusted.

7. Code Link on Git

All analysis code is maintained at:

`https://github.com/omnomnomm/dfhr_graph_analysis`

The repository is organised by stage rather than by variant, so a given step runs across all three systems without duplicated code. `notebooks/PCN_Construction.ipynb` carries the full pipeline from coordinate extraction through to the hybrid networks and diagnostics, with the variant list set at the top so any subset can be rebuilt. `scripts/` holds the GROMACS commands for trajectory fitting, reference frame extraction and RMSD. `analysis/networks/` receives the pickled NetworkX graph objects, one file per variant and network type, each holding ten windows. `figures/` receives the output figures in one subdirectory per notebook section.

The notebook caches intermediate products, so a rerun does not repeat the expensive steps. Coordinate extraction takes around 140 seconds per variant and is cached as a NumPy array, and occupancy matrices are cached per window. Everything after that is fast enough to rerun freely while varying parameters. Raw trajectories are not in the repository, since they run to tens of gigabytes per variant with water retained; paths to them are set in the configuration cell at the top of the notebook.

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